

4.4B

Data Modeling in SAP Analytics Cloud¹

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OBJECTIVE

To acquire and model data for analysis using the "new" modeler in SAP Analytics Cloud.

DATA REQUIRED

- [GB_AnalyticsData3.xlsx](#)

Scenario

The entire sales force from Global Bike will be meeting in a couple weeks for their annual retreat where they perform various team building activities, enjoy time away from the office, and more importantly, examine past sales performance to strategize and plan for the future. VP of Sales, Nina Kane has asked you to prepare some historical data so that her business analysts can prepare for the retreat. Therefore, you have downloaded some sales data from the ERP system for this purpose.

The following is an Entity-relationship diagram of the data downloaded to Excel.

¹ This exercise is based on the Global Bike data model developed collaboratively by Klaus Fryburger, Tobias Hagen, and Nancy Jones 2019 and provided as a public model on the UA SAC academic tenant.

ER Model

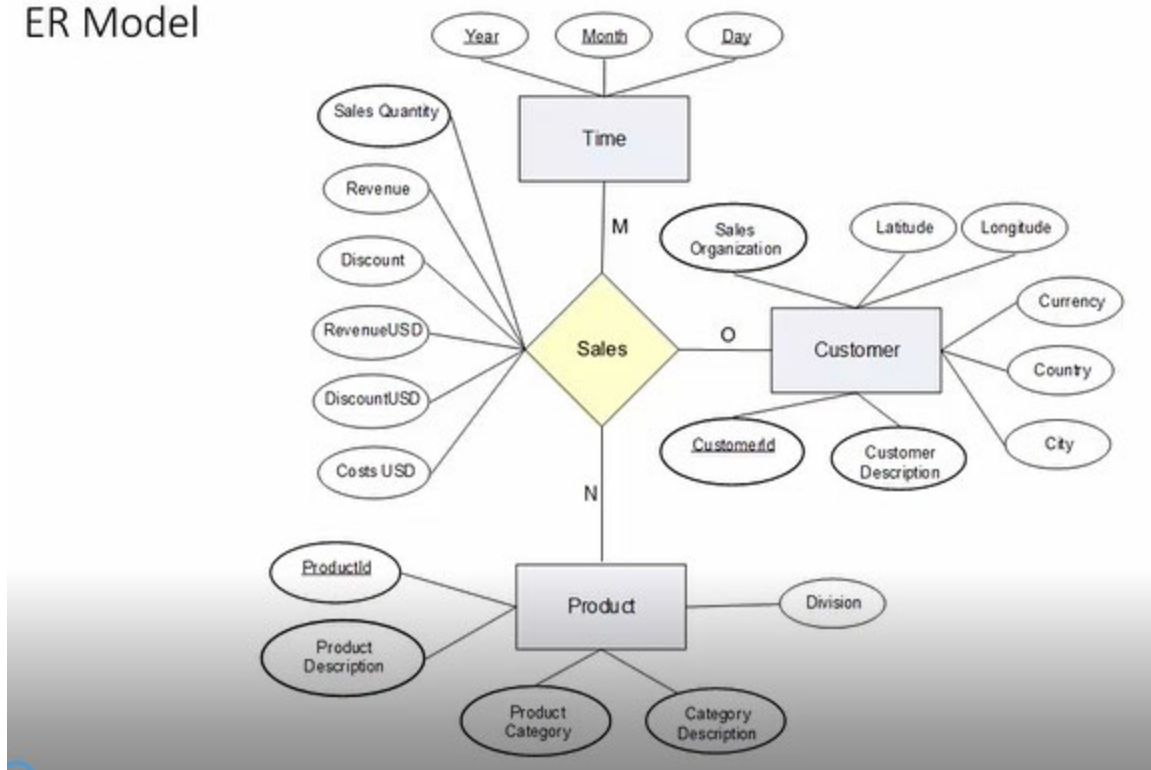


Figure 1: ERD of Global Bike Sales Data created by Dr. Klaus Fryburger, Ludwigshafen University

Background

Data for analysis may be acquired via various means and from various sources. All acquired data should first be examined for anomalies and inconsistencies and any issues addressed before the analysis is performed. This process of data transformation, sometimes called “data wrangling”, is imperative to mitigate the effects of “GIGO”, garbage-in-garbage-out and to provide opportunities for robust analyses.

In SAP Analytics Cloud (SAC), we create data models from imported or linked data sources. The models can consist of one or many data sets and it is in the modeling process that we can identify and fix/transform any data inconsistencies, add hierarchical structures, filter data, merge data sets, create calculated columns, and otherwise prepare the data for analysis.

SAC allows us to create models in various ways. A “public” model is created to be shared and is subject to corporate governance such as access controls or quality control. A public model can be used by many different analyses and/or many different users. A “private” or “embedded” model is one that is contained entirely in a Story. Although it is not shareable as a model, the story can be shared. We would create private model for an ad hoc analysis or when governance is not required. A

third type of model is called a “Dataset”. It is shareable and contains fewer features than a Model. It is primarily used for predictive scenarios.

There are two types of Models in SAC – the classic account model and the “new model”. The classic account model stores measures as an account dimension in a star schema. The new model stores measures as separate entities. There are pros and cons to each, but generally speaking, the classic model is simpler to build. This assignment uses a classic account model.

Creating the Model

1. In SAP Analytics Cloud (SAC), go to Modeler.
2. Select Create New Model
3. Select Start with data, Next
4. Choose File (Local File or File Server)
5. At the Create Model from File prompt, click on Select Source file and choose the Excel file [*GB_AnalyticsData3.xlsx*](#).
 - a. Use first row as column headers should be selected.
6. Import. This is a large file so it may take some time.
7. Notice that SAC identified numerical variables as measures although some should be dimensions. To change the misidentified variables do the following.
 - a. Select the checkbox to the left of *Latitude, Longitude, OrderNumber, OrderItem, Year, Month, Day, and Customer*.
 - b. Select Convert to Dimension as shown in Figure 2.

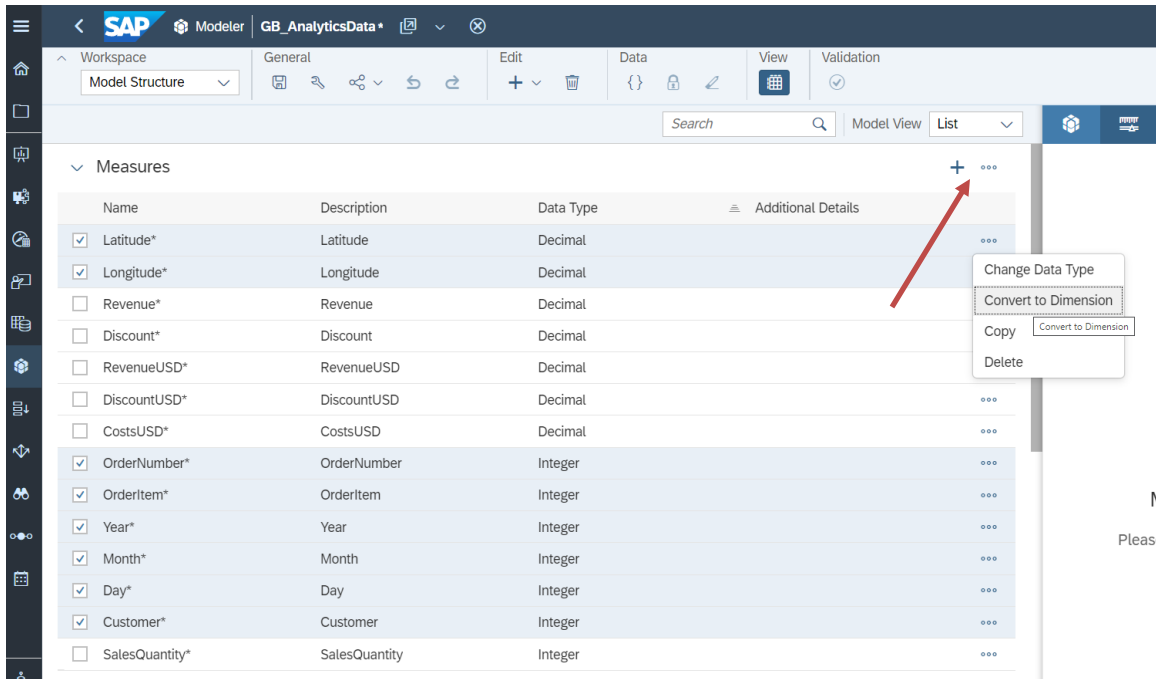


Figure 2: Select the Mis-identified Measures

- c. Only *Revenue*, *Discount*, *RevenueUSD*, *DiscountUSD*, *CostsUSD*, and *SalesQuantity* should remain as Measures.
8. Deselect the new dimensions.
9. Examine *Customer* and the other variables in the dataset. Notice that some of the dimension variables are descriptions or attributes of Customer. We can “link” these related variables using the Properties of the modeler dimension details.
 - (1) Select *Customer* either by checking the box in the Dimension list or by clicking on the Customer heading of the Data Foundation panel. *Hint: If you don't see the Data Foundation panel, click on View in the top ribbon.* You will now see Customer dimension details in a panel to the right of the screen. If you do not see the details, click on Details in the upper right of the modeling panel, (shown in Figure 3).

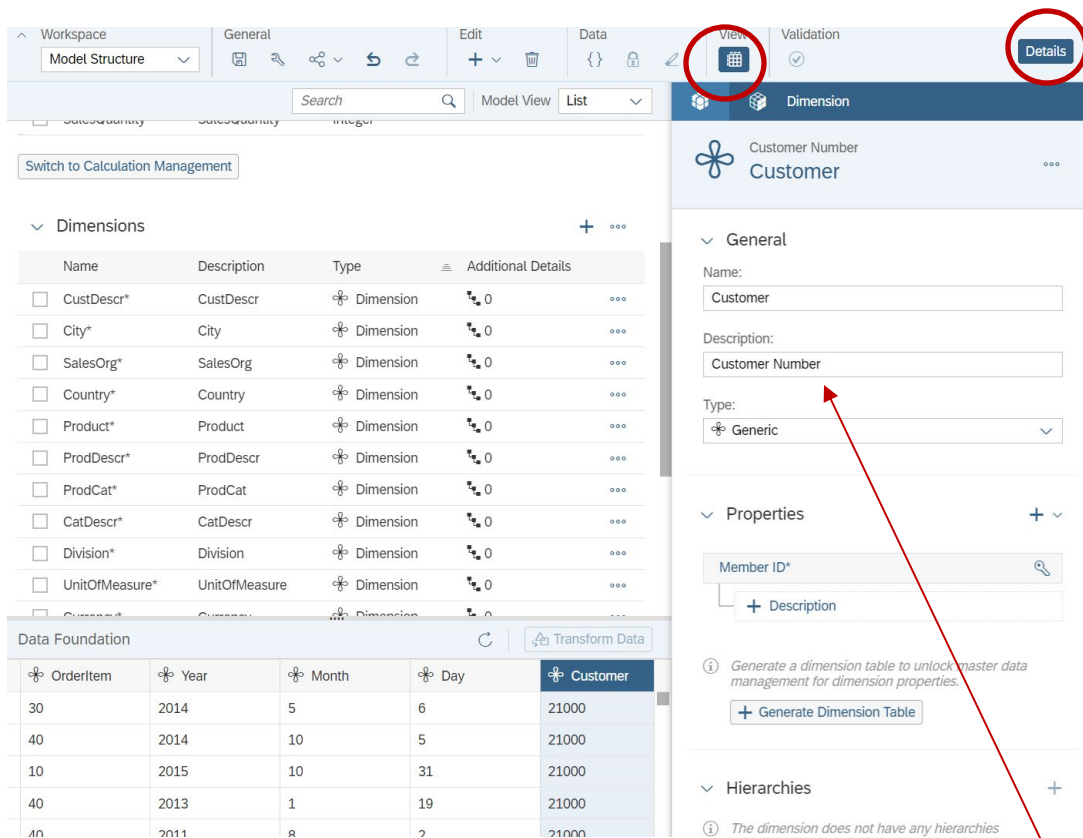


Figure 3: Dimension Attributes

- (i) Type the *Customer* variable description in the data table as “Customer Number”.
- (ii) Under Properties, add the customer description (name) by selecting + Description and selecting *CustDescr* from the dropdown list. OK. Now the customer number and the customer’s name are linked.
- (iii) Continue to add Properties to the Customer dimension by selecting the + sign to the right of Properties and From Source.

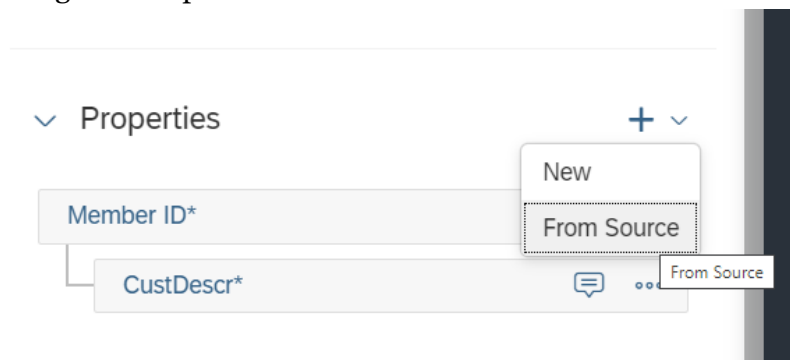


Figure 4: Adding Properties to a Dimension

(iv) Use the dropdown list to select each of the following six attributes one at a time. When you are finished your *Customer* dimension details will look like Figure 5.

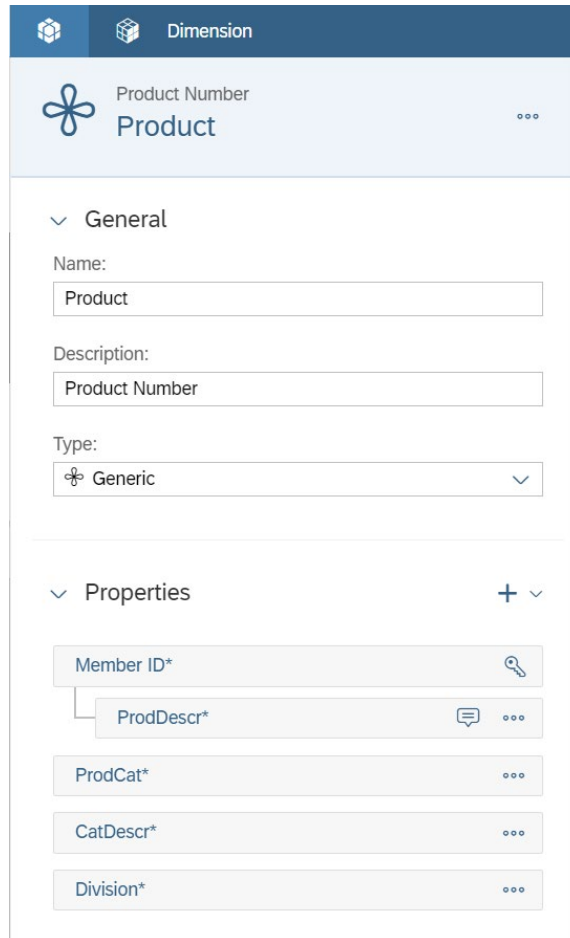
- Add Property *City*.
- Add Property *SalesOrg*.
- Add Property *Country*.
- Add Property *Currency*.
- Add Property *Latitude*.
- Add Property *Longitude*.

The screenshot shows a web interface for configuring a dimension. At the top, there's a blue header with a cube icon and the word "Dimension". Below this is a light blue section with a flower icon, "Customer Number", and "Customer". A dropdown menu is open, showing "Generic" selected. Below this is a "General" section with fields for "Name" (Customer), "Description" (Customer Number), and "Type" (Generic). Below the "General" section is a "Properties" section with a list of attributes: "Member ID*", "CustDescr*", "City*", "SalesOrg*", "Country*", "Currency*", "Latitude*", and "Longitude*". Each attribute has a key icon or a dropdown menu.

Figure 5: Customer Properties

10. Examine *Product* and add attributes. Use the same process you used for *Customer* to add properties to *Product* as detailed below. When you are finished your *Product* dimension details will look like Figure 6.

- a. Type the *Product* variable description in the data table as “Product Number”.
- b. Add the following Dimension Attributes to *Product*.
 - (1) Add Description *ProdDescr*.
 - (2) Add Property *ProdCat*.
 - (3) Add Property *CatDescr*.
 - (4) Add Property *Division*.



The screenshot displays the configuration interface for a dimension named 'Product'. The 'General' section includes fields for 'Name' (set to 'Product'), 'Description' (set to 'Product Number'), and 'Type' (set to 'Generic'). The 'Properties' section is expanded, showing a list of attributes: 'Member ID*' (marked as a key), 'ProdDescr*' (with a comment icon), 'ProdcCat*', 'CatDescr*', and 'Division*', each with a three-dot menu icon.

Figure 6: Product Attributes

11. To further enhance the Customer and Product dimensions, we can add a hierarchy which allows the user to look at the data at different levels of granularity or detail.
To create a Customer Hierarchy:
 - a. Select the *Customer* column.
 - b. At the bottom of the Details panel, choose the + to the right of Hierarchies or + Level-Based Hierarchy to create a hierarchy.

The screenshot shows the 'Data Foundation' interface. On the left, a table is displayed with the following data:

Member ID	AA CustDescr	AA City	AA SalesOrg	AA Country
21000	Ostseerad	Anklam	DN00	DE
21000	Ostseerad	Anklam	DN00	DE
21000	Ostseerad	Anklam	DN00	DE
21000	Ostseerad	Anklam	DN00	DE
21000	Ostseerad	Anklam	DN00	DE
21000	Ostseerad	Anklam	DN00	DE
21000	Ostseerad	Anklam	DN00	DE
21000	Ostseerad	Anklam	DN00	DE
21000	Ostseerad	Anklam	DN00	DE

On the right, there are options to 'Generate a dimension table to unlock master data management for dimension properties.' with a '+ Generate Dimension Table' button. Below this, under 'Hierarchies', there is a message: 'The dimension does not have any hierarchies maintained. Create hierarchies to logically organize your data.' with a '+ Level-Based Hierarchy' button. A 'Create Hierarchy' button is also visible on the far right.

Figure 7: Create Hierarchy

- c. Type "CustomerHierarchy" as the ID.
- d. Type "Customer Hierarchy" as the Description.
- e. The *Member ID* (Customer) will default as level 1 of the hierarchy because there are no other levels in the hierarchy yet.
 - (1) Select the more actions next to *Country* and set as level 1.
 - (2) Select the more actions next to *SalesOrg* and set as level 2.
- f. The results should look like Figure 8. Member ID is the most detailed level and Country is the least detailed level of this hierarchy.

The screenshot shows the 'Hierarchy Builder' window. On the left, a list of properties is available: City, SalesOrg, Country, Currency, Latitude, Longitude, and Member ID. The 'Member ID' property is marked with a key icon, indicating it is the primary key. On the right, the 'CustomerHierarchy' is being configured. The 'ID' field is set to 'CustomerHierarchy' and the 'Description' field is set to 'Customer Hierarchy'. Below these fields, the 'Levels' section shows a hierarchy with three levels: Level 1 is 'Country', Level 2 is 'SalesOrg', and Level 3 is 'Member ID'. Each level is marked with a checkmark and a level number. At the bottom right, there are 'OK' and 'Cancel' buttons.

Figure 8: Customer Hierarchy

- g. OK.
12. Add a Product Hierarchy. The process is the same as for the Customer Hierarchy.
- Select the *Product* column.
 - On the Hierarchy Builder, click on the +.
 - Name the hierarchy ID “ProductHierarchy”.
 - Type “Product Hierarchy” as the Description.
 - Add levels to the hierarchy as follows:
 - Set *Division* as level 1.
 - Set *CatDescr* as level 2.
 - The results should look like Figure 9.

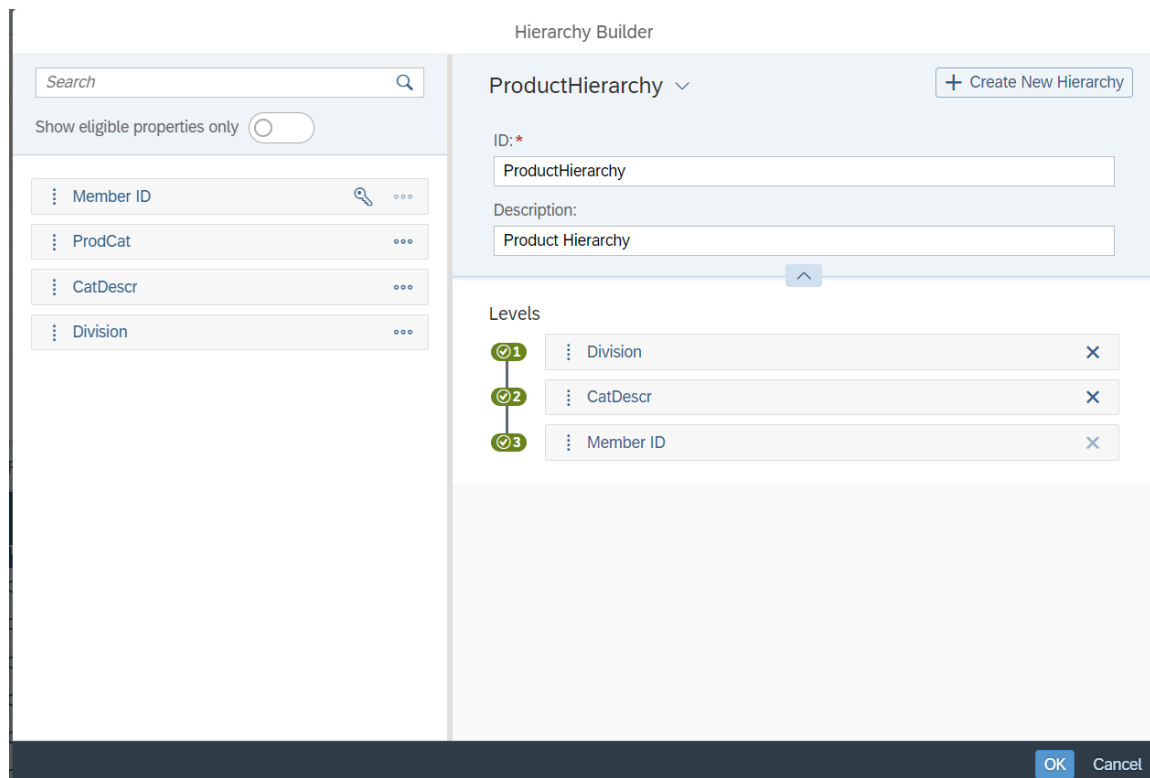


Figure 9: Product Hierarchy

g. OK.

13. Since *UnitOfMeasure* contains only one value, “ST” (stücke in German and each or piece in English), we can remove the unit of measure column.
- Select the *UnitOfMeasure* column.
 - On the More Actions dropdown, choose Delete. Be careful to delete only *UnitOfMeasure*.

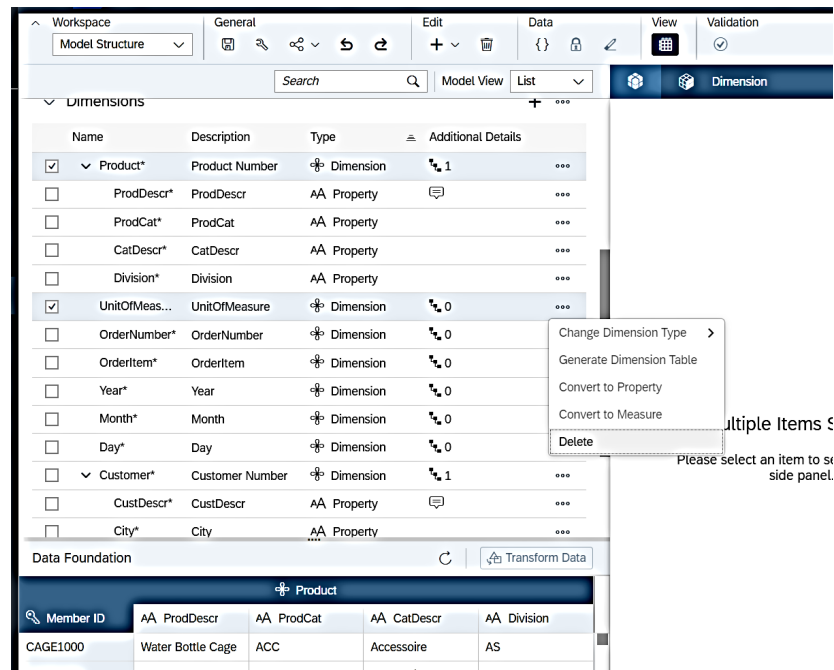


Figure 10: More Actions for a Column

14. Add a geo enrichment based on latitude and longitude by customer.

- a. Select *Longitude* from the Dimension list.
- b. Under the Dimension Property of the Details panel, select the pencil in the Type field to edit the Dimension type.
 - (1) From the dropdown menu select Longitude to indicate that this dimension is actually the geo-dimension of longitude.

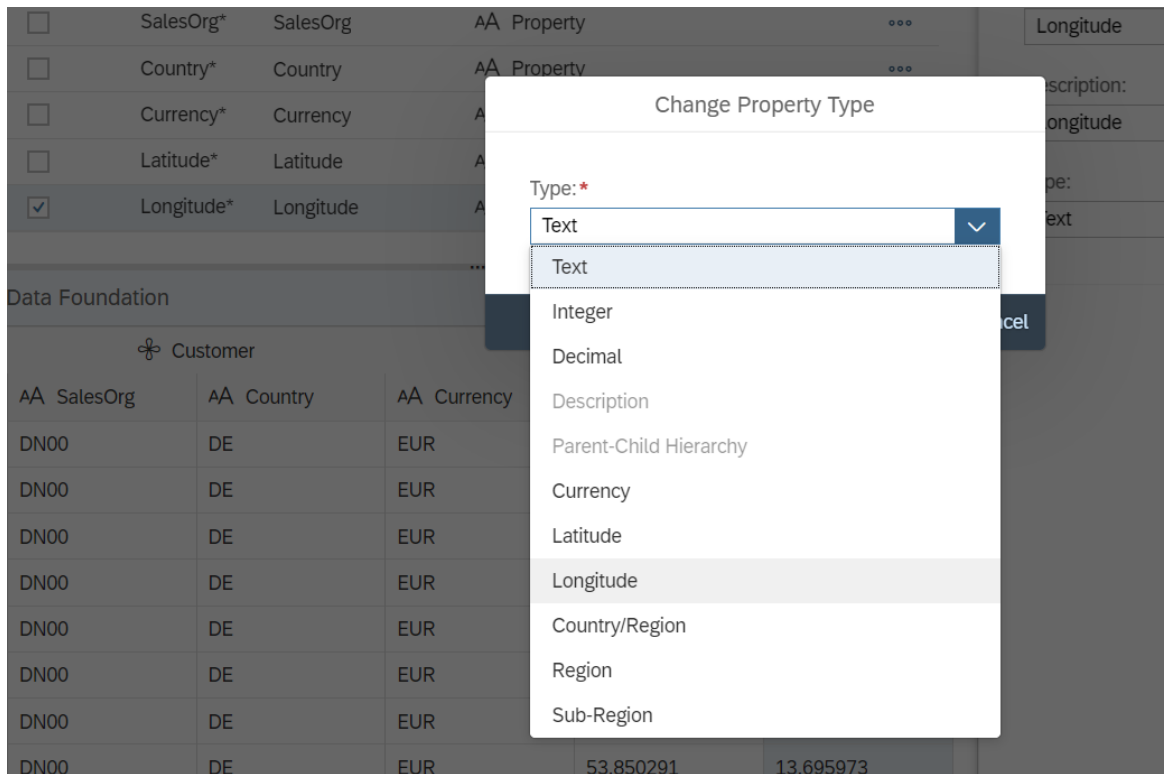


Figure 11: Create a geo-dimension

- c. Next. You will see that Longitude is now indicated as a geo-dimension.
- d. Use the more actions next to *Latitude* to change the property type to Latitude. OK. OK.
- e. You will receive an error that there are some validation errors in the name of some properties. Click on Review. You will see that Currency is a reserved keyword.

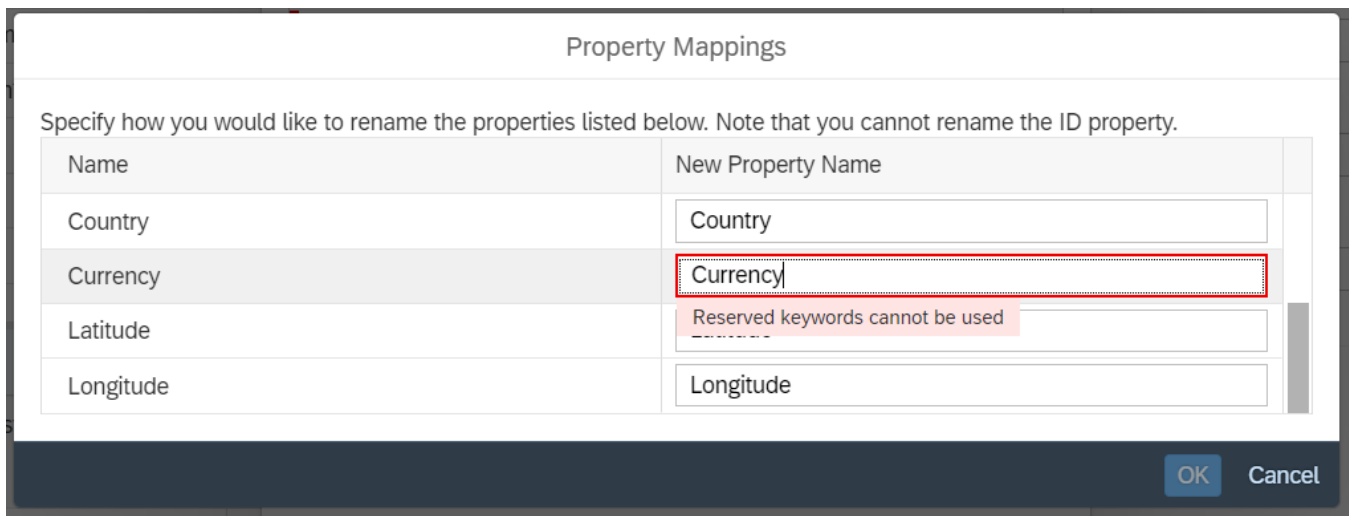


Figure 12: Data Validation Error

- f. Type “RegionalCurrency” as the New Property Name for Currency. *Hint: spaces are not allowed in the name of the property.*
- g. OK. Customer will be converted to a dimension table.

15. Select *Product*.

- a. At the bottom of the details panel select + Generate Dimension Table. OK. You will now see that both Customer and Product are Dimension tables.

▼ Dimensions

Name	Description	Type	Additional Details
☐ Product*	Product Number	Product*	30 1
☐ Customer*	Customer Number	Customer*	25 1
☐ OrderNumber*	OrderNumber	Dimension	0
☐ OrderItem*	OrderItem	Dimension	0
☐ Year*	Year	Dimension	0
☐ Month*	Month	Dimension	0
☐ Day*	Day	Dimension	0

Figure 13: Dimension Tables

- 16. Save. Save the model to your assigned folder. I named mine “Global Bikes”.
- 17. Notice that the transaction dates are stored as separate dimension variables: year, month and day. To form a date column, we will need to combine (or concatenate) the three dimensions.
 - a. Select at least one of the columns *Year*, *Month*, and *Day*.
 - b. Click on Transform Data on the Data Foundation panel ribbon as shown in Figure 14. This will take you to a different screen where you can transform your data.

Show Unpublish... OFF Actual

Year	Month	Day	22	SalesQuantity	1 ²³
2022	7	18	0		0
2021	4	10	0		0
2022	5	14	0		0
2022	6	21	0		0
2022	7	23	0		0

Figure 14: Transform Data

- c. This will take you to a different screen where you can transform your data. You will also receive a message that states that only a sample of the data will be shown. Acknowledge the message by clicking on Got it!

- d. Select the columns clicking on the headers Year, Month, and Day and using the shift key.
- (1) On the column actions, choose Create a Transformation.

	AA OrderNu...	AA OrderItem	AA Year	AA Month	AA Day	22 SalesQua...	123 Revenue	123 Disc
1922	200040	10	2007	1	29			640
1921	200035	100	2007	1	26			1.8
1920	200035	90	2007	1	26	2		320
565	206161	50	2011	1	24	1		172.11
564	206156	40	2011	1	20	1	3442.17	172.11

Figure 15: Create Column Transformation

- (2) On the Create Transformation dropdown, choose Concatenate.
- (3) Add / to the delimiter between the “” in the transformation window as shown in Figure 17. *Hints: Just type where it says value as shown in Figure 16. Be careful to not add any extra spaces between the “”.*

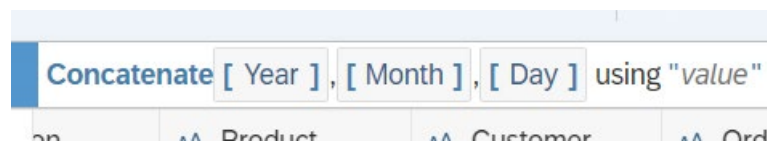


Figure 16: Concatenate Prompt

	AA OrderNu...	AA OrderItem	AA Year	AA Month	AA Day	AA Year_Mon...	22 SalesQua...	123 Rever
	100074	20	2007	2	16	16	8	186.74
	100311	40	2007	4	23	23	4	93.36

Figure 17: Concatenating Year, Month and Day

- (i) Select the check mark to accept the transformation.
- (4) The three columns will be merged into one. In the Details panel for the new column make the following changes.
- (i) Change the column to data type Date.
 - (ii) Change the name of the column to “Date”.
 - (iii) Add a Description “Transaction Date”.

The screenshot shows a data transformation interface. On the left, a table with columns 'Day', 'Date', and 'SalesQuantity' is displayed. A context menu is open over the 'Date' column. On the right, a 'Details' panel for the 'Date' field is shown. The 'General' section has 'Name' set to 'Date' and 'Description' set to 'Transaction Date'. The 'Dimension Table' section states 'This dimension does not use a dimension table.' The 'Data' section has 'Data Type' set to 'Date' and 'Conversion Format' set to 'YYYY/MM/DD'. Below this, there are tabs for 'Data Distribution' and 'Validation'. The 'Data Distribution' tab is active, showing 'Sampled Rows: 2000 / 171010', 'Unique Values: 605', and a bar chart for 'Group Values' with a count of 18 for the date '2007/6/11'.

Figure 18: Changing the Data Type

18. Close Data Transformation. Save Changes before closing. You will be returned to the modeler workspace.

19. “Finetuning” the Measures.

a. Select *Revenue* from the list of Measures.

(1) Under Data Type change the decimals to 2 by clicking on the edit pencil and selecting 2 from the dropdown menu. OK.

(2) Under Units & Currencies select Unit Type Currency. *Special note: the default will be the USD sign for the currency. We will have to rely on the RegionalCurrency value to tell us if the amounts are dollars or euros.*

b. Create the same settings for *Discount*, *RevenueUSD*, *DiscountUSD* and *CostsUSD*.

c. Select *SalesQuantity* from the list of Measures.

(1) Under Units & Currencies select Unit Type Unit.

(i) Type “EA” into the Fixed Unit field.

d. Now you can view your completed model as a graph under Model View.

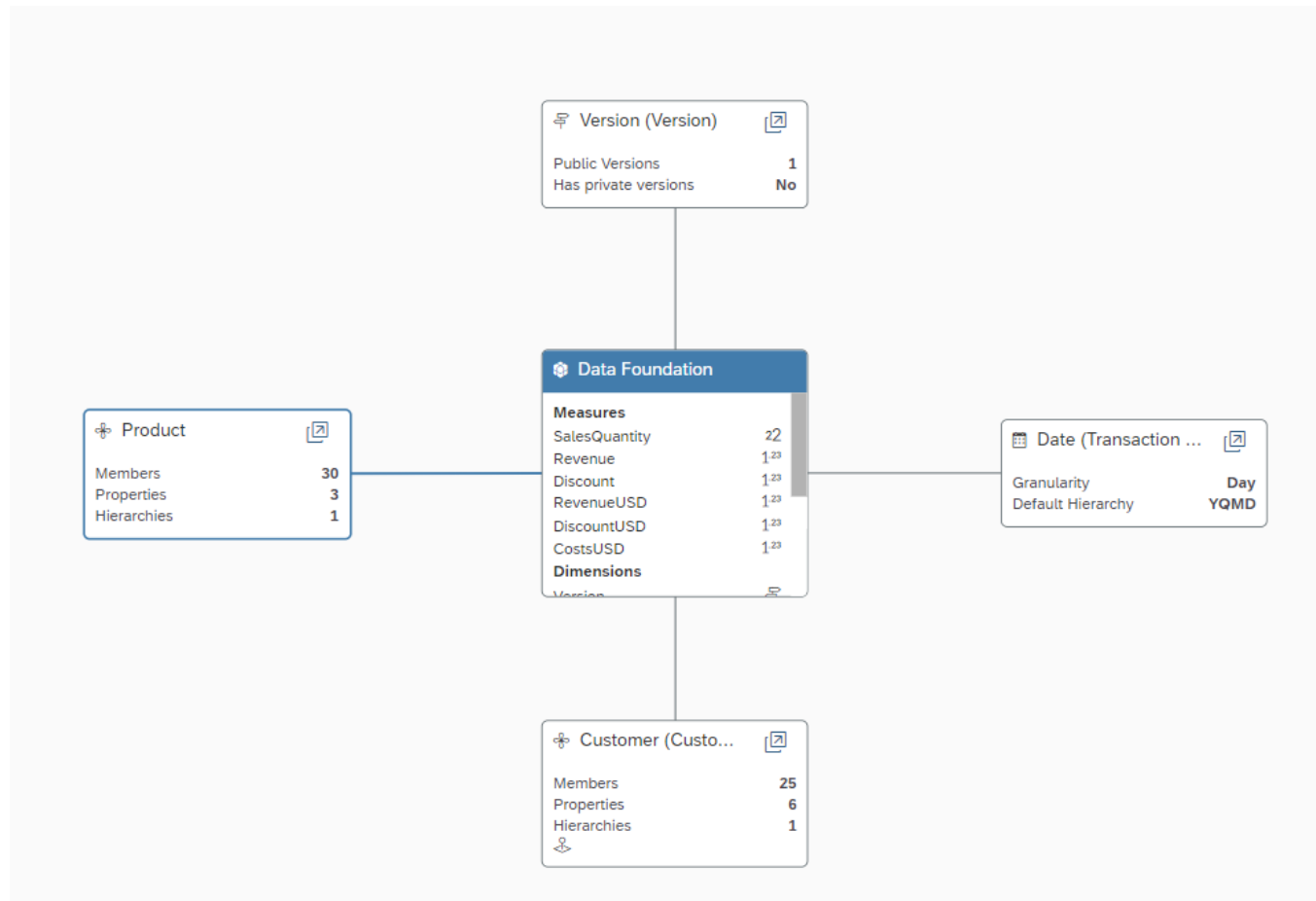


Figure 19: The Model View of Global Bikes

20. Save.

The Global Bike sales data model is now ready and available for analysis. It is considered a public model and may be used by others for their analyses.

21. Share your data model with user **Your Instructor** and provide Full Control. The Share icon is on the top ribbon under General.

- Do NOT send an email.
- Copy the file link to the clip board to paste into the assignment submission in your learning management system.

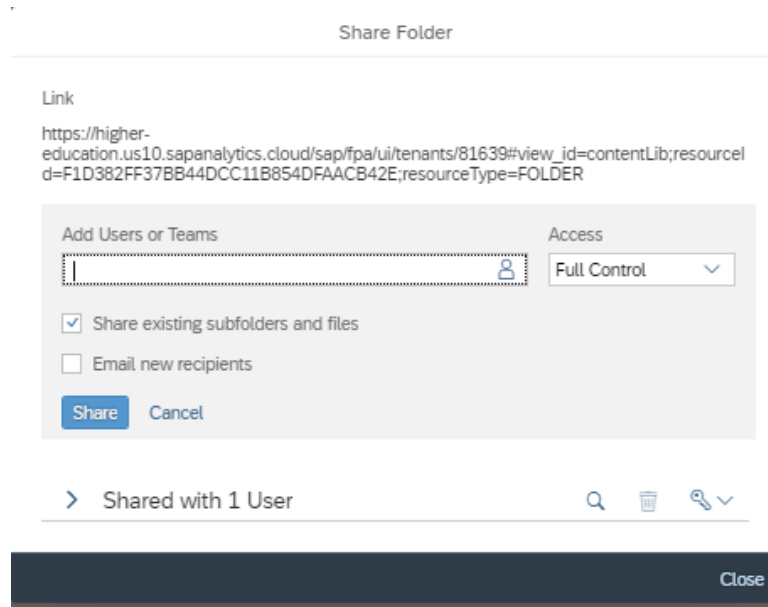


Figure 20: Sharing the Folder and Copying the Link

SUBMISSION FOR GRADING

To submit this assignment for grading, simply type the name of your model (example: *Jones_GlobalBike*) and paste the folder link (from step 21b) into your assignment submission. It will be graded in SAC directly.